

Population III Stars

- First stars in Universe ($z \sim 15-30$)
- Formed from pristine H/He gas
- Key characteristics:
 - Massive (10s-1000s $M_{\odot}$)
 - Short-lived ($\sim$ Myr timescales)
 - Fundamental for reionization and metal enrichment

Why Cosmology?

- Isolated collapse issues:
 - Simplistic model
 - Tidal forces from large-scale structure matter
 - Realistic angular momentum acquisition requires cosmological context
- Our approach: Full cosmological initial conditions
 - Planck cosmology parameters (observations = good)
 - Includes dark matter + baryons
 - Self-consistent structure formation
 - Large volumes allow to explore statistics and time evolution

Zoom-in Simulations: Why We Need Them

- Cosmic variance/gas accretion for star formation → Need large volumes
- But resolve Pop III physics → High resolution
- Solution: Hierarchical zoom-in
 - Large parent box (100 Mpc/h?)
 - Initial progressive refinement to $\sim$ kpc scales
 - Particle splitting to go to $\sim$ pc scales

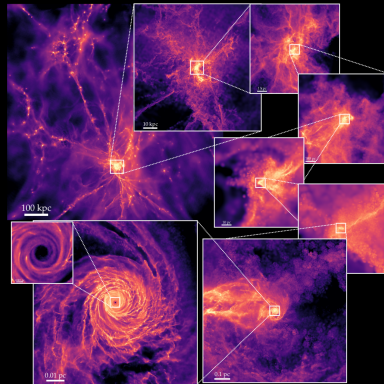


Figure: STARFORGE + FIRE
Zoom-in. Credit: 2309.13115

Multi-Scale Initial Conditions (MUSIC)

Steps for zoom-in ICs refinement with MUSIC:

- Set up lowest-res grid
- Pick a region of interest for further refinement (for us, based on generated halo tables)
- Increase particle density by a factor of 2 for every next region
- Generate density field and convolve it with previous level
- Baryons are only placed in the highest-resolution region

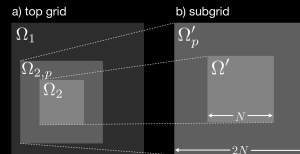
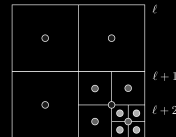


Figure: MUSIC zoom-in grid structure. Credit: Hahn, Abel, 2011

High-redshift Halo Finding

- Key challenges at target redshift $z \gtrsim 15$:
 - Low particle counts in early halos (Fixed by decreasing box size)
 - Halo finders are mostly calibrated at low z
 - Numerical artifacts (Current bottleneck for $z > 13$)
- Calibration approaches:
 - MUSIC $\rightarrow$ Hydro $\rightarrow$ Halo finder (accurate but slow, ~ 20 hrs)
 - MUSIC $\rightarrow$ **PeakPatch** (fast semi-analytic, ~ 30 min)

What is PeakPatch?

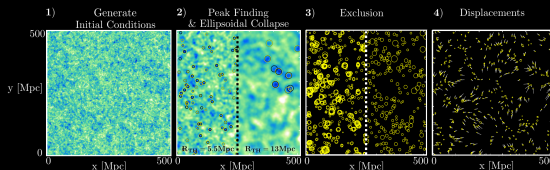
Semi-analytic halo finder based on peak patches

● Key features:

- Fast → enables large parameter surveys
- Works at any redshift (in theory)
- Lightcones!

● Main steps of the code:

- Looks at different filter scales
- Resolves homogeneous ellipsoidal collapse
- Excludes overlapping halos
- Performs lagrangian displacements after peaks were found



PeakPatch algorithm overview. Credit: Stein et al., 2018

PeakPatch Tiling Strategy

Domain decomposition into tiles

- Each tile has:
 - Core region
 - Buffer zones (n_{buff} layers)
- $R_{\text{smooth}}^{\text{max}}$ sets maximum halo radius → **should be less than n_{buff}**
- Buffer size ensures boundary independence

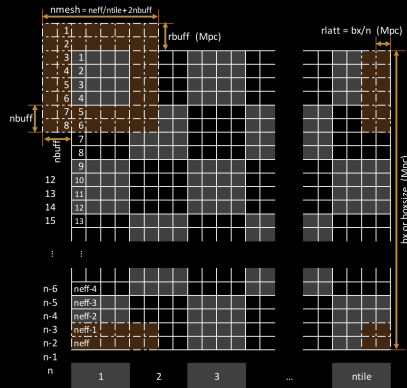


Figure: PeakPatch tile structure with buffer zones

Extending PeakPatch to higher redshift

Why not just change redshift of the PeakPatch run?

- Will find 0 halos at high z – large halos won't have time to form
- **Decrease $R_{\text{smooth}}^{\text{max}}$** – assuming it changes with z
- Simply decreasing $R_{\text{smooth}}^{\text{max}}$ would eventually yield $R_{\text{smooth}}^{\text{max}} < a_{\text{cell}}$ (size of the cell)
- If also **decrease the box size**, now we're good!
- **Keep $R_{\text{smooth}}^{\text{max}}/L_{\text{boxsize}}$ constant to avoid buffer issues!**
- There *might be issues* with this approach, more on that later...

$$R^{\text{max}}(z) \approx R_0^{\text{max}} e^{-\kappa z} \quad c$$

$$\kappa = -\frac{1}{z_j} \ln \left(\frac{R_j^{\text{max}}}{R_0^{\text{max}}} \right)$$

$$z_j = 5, R_j^{\text{max}} = 2 \text{ Mpc/h}$$

$$R_0^{\text{max}} = 34 \text{ Mpc/h}$$

Power law formula we use for approximating max halo size at low redshifts (breaks down at higher z)

Halo Mass Function, Transfer Function Results

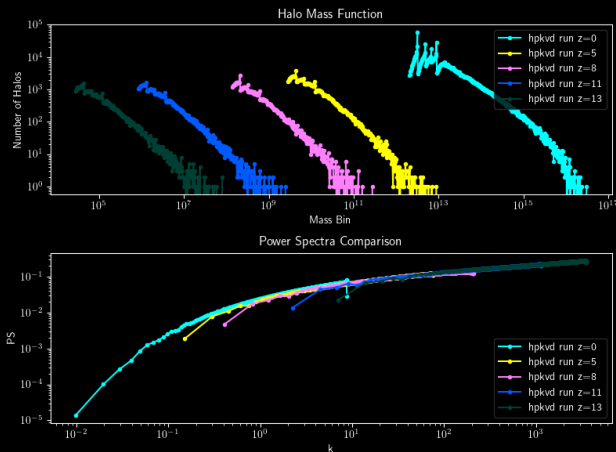


Figure: Halo Mass Functions and Transfer Functions

Output fields/halos

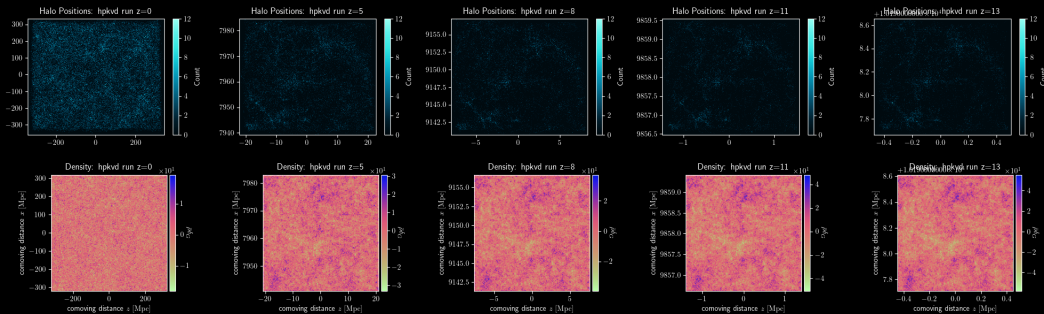


Figure: Halos detected and initial overdensity field

It seems to be working!..

Surprisingly, this approach works for $z \lesssim 20$!

- We get **1s** of **low-k** (high-R) halos, and $10^3 - 10^4$ high-k (low-R) halos for $z \lesssim \mathbf{17}$ (good)
- For $z \sim \mathbf{20}$, we get **0** halos at **low-k** and $10^2 - 10^3$ halos at **high-k**, which means that **the formula overestimates the largest halo size** at high- z

$$R^{\max}(z) \approx R_0^{\max} e^{-\kappa z} \quad c$$

$$\kappa = -\frac{1}{z_j} \ln \left(\frac{R_j^{\max}}{R_0^{\max}} \right)$$

$$z_j = 5, R_j^{\max} = 2 \text{ Mpc/h}$$

$$R_0^{\max} = 34 \text{ Mpc/h}$$

... but maybe it doesn't?

This is actually problematic and might be unphysical...

- Boxsize at $z \sim 8$ is ~ 10 Mpc, with starting $z = 0$ boxsize of 1200 Mpc!
- Boxsize at $z \sim 20$ is ~ 10 kpc, which is **too small**

$$R^{\max}(z) \approx R_0^{\max} e^{-\kappa z} \quad c$$

$$\kappa = -\frac{1}{z_j} \ln \left(\frac{R_j^{\max}}{R_0^{\max}} \right)$$

$$z_j = 5, R_j^{\max} = 2 \text{ Mpc}/h$$

$$R_0^{\max} = 34 \text{ Mpc}/h$$

Current pipeline status

- MUSIC (DM-only ICs)
- PeakPatch (Halo finding)*
- Halo choice
- MUSIC (Zoom-in DM+baryons)
- Hydro (cosmological, up to high- z)
- Hydro (FIRE, non-cosmological)
- Hydro (FIRE+STARFORGE, non-cosmological)

* A bug in PeakPatch currently is preventing us to go to redshifts higher than ~ 13 !

The future looks patchy, but the stars are aligning!

Other directions we can go with:

- Getting **PopIII IMF** (initial goal)
- **PINGs** (Primordial Intermittent Non-Gaussianities)
- Evolution of a halo at **different redshifts**
- Using **PeakPatch** for halo finding on **hierarchical MUSIC grids**
- PeakPatch as **ICs generator** (VERY down the line)

Key Takeaways

- **Goal:** Study Pop III star formation with **full cosmological simulations**, combining:
 - **MUSIC** (multi-scale zoom-in ICs)
 - **PeakPatch** (fast semi-analytic halo finding)
 - **GIZMO/FIRE/STARFORGE** (hydrodynamics + stellar feedback)
- **Successes:**
 - Validated hierarchical zoom-in approach for $z \lesssim 13$
 - Achieved halo detection at high- z with PeakPatch
- **Challenges:**
 - High- z halo finder calibration ($z > 13$ bottleneck)
 - Unphysical (?) box sizes
- **Next Steps:**
 - Debug high- z PeakPatch implementation
 - Integrate hydro simulations with cosmological ICs
 - Pursue Pop III IMF and (potentially) primordial non-Gaussianities (PINGs)

Halo finding output in PeakPatch

z=8

Box 3 of 8

D = 0.144

z = 7.83

Npks = 8494

Smoothing:

R = 0.365 Mpc: sigma = 3.002	Peaks found = 0
R = 0.317 Mpc: sigma = 3.145	Peaks found = 3
R = 0.276 Mpc: sigma = 3.289	Peaks found = 2
R = 0.240 Mpc: sigma = 3.430	Peaks found = 5
R = 0.209 Mpc: sigma = 3.569	Peaks found = 13
R = 0.181 Mpc: sigma = 3.713	Peaks found = 41
R = 0.158 Mpc: sigma = 3.860	Peaks found = 85
R = 0.137 Mpc: sigma = 4.009	Peaks found = 147
R = 0.119 Mpc: sigma = 4.160	Peaks found = 267
R = 0.104 Mpc: sigma = 4.313	Peaks found = 436
R = 0.090 Mpc: sigma = 4.467	Peaks found = 706
R = 0.078 Mpc: sigma = 4.622	Peaks found = 1080
R = 0.068 Mpc: sigma = 4.776	Peaks found = 1680
R = 0.059 Mpc: sigma = 4.924	Peaks found = 2441

z=20

Box 1 of 8

D = 0.006

z = *****

Npks = 0

Smoothing:

R = 0.000 Mpc: sigma = 8.702	Peaks found = 0
R = 0.000 Mpc: sigma = 8.967	Peaks found = 0
R = 0.000 Mpc: sigma = 9.437	Peaks found = 0
R = 0.000 Mpc: sigma = 9.893	Peaks found = 0
R = 0.000 Mpc: sigma = *****	Peaks found = 0
R = 0.000 Mpc: sigma = *****	Peaks found = 0
R = 0.000 Mpc: sigma = *****	Peaks found = 0
R = 0.000 Mpc: sigma = *****	Peaks found = 0
R = 0.000 Mpc: sigma = *****	Peaks found = 2
R = 0.000 Mpc: sigma = *****	Peaks found = 4
R = 0.000 Mpc: sigma = *****	Peaks found = 9
R = 0.000 Mpc: sigma = *****	Peaks found = 22
R = 0.000 Mpc: sigma = *****	Peaks found = 51
R = 0.000 Mpc: sigma = *****	Peaks found = 87
R = 0.000 Mpc: sigma = *****	Peaks found = 130
R = 0.000 Mpc: sigma = *****	Peaks found = 198
R = 0.000 Mpc: sigma = *****	Peaks found = 349
R = 0.000 Mpc: sigma = *****	Peaks found = 532
R = 0.000 Mpc: sigma = *****	Peaks found = 757
R = 0.000 Mpc: sigma = *****	Peaks found = 1192
R = 0.000 Mpc: sigma = *****	Peaks found = 1733